

## 14 Normal Modes pt 2

We have a matrix representing the normal modes of our system

$$\mathbb{N}(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} = \begin{pmatrix} A_+ \cos(\omega_+ t) \\ A_- \cos(\omega_- t) \end{pmatrix}$$

It is nice to express this as a complex function, but where we could still recover this. Cos is the real component of  $e^{i\omega t}$ , so we write the complex mode matrix

$$\mathbb{N}' = \begin{pmatrix} C_+ e^{i\omega_+ t} \\ C_- e^{i\omega_- t} \end{pmatrix}$$

Where  $C_+ = A_+ e^{i\delta t}$ , and  $C_- = A_- e^{i\delta t}$ .

We rewrite  $\mathbb{N}'(t)$  as

$$\mathbb{N}'(t) = C_+ \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^{i\omega_+ t} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{i\omega_- t}$$

where these functions are clearly just the basis functions.

For our original variables, we can write out  $\mathbb{Z}(t)$ .

$$\mathbb{Z}(t) = \begin{pmatrix} z_1(t) \\ z_2(t) \end{pmatrix} = \frac{C_+}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{i\omega_+ t} + \frac{C_-}{2} \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{i\omega_- t}$$

To go back to our actual values, we just have

$$\mathbb{X}(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} = \text{Re}(\mathbb{Z}(t))$$

What do we do when we can't guess the normal modes?

### Matrix approach

Our goal is to write the E-L equations as a vector equation, and solve for the eigenvectors.

Lets take our E-L equations from before

$$\begin{aligned} \ddot{x}_1 &= -kx_1 - k'x_1 + k'x_2 \\ \ddot{x}_2 &= -kx_2 - k'x_2 + k'x_1 \\ m \begin{pmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{pmatrix} &= - \begin{pmatrix} k + k' - k' \\ -k' + k + k' \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \end{aligned}$$

We can rewrite this in a nicer matrix way

$$\ddot{\mathbb{X}} = -\frac{1}{m} \mathbb{K} \mathbb{X}$$

We can solve this by guessing through an Ansatz.

Lets look at the slightly more complicated but nicer matrix  $\mathbb{Z}$ .

$$\ddot{\mathbb{Z}} = -\frac{1}{m} \mathbb{K} \mathbb{Z}$$

With the Ansatz  $\mathbb{Z}(t) = \underbrace{\mathbb{Z}_0}_{\text{constant from init conds}} e^{i\omega t}$

We have

$$\begin{aligned}\frac{d^2}{dt^2}(\mathbb{Z}_0 e^{i\omega t}) &= -\frac{1}{m} \mathbb{K} \mathbb{Z}_0 e^{i\omega t} \\ -\omega^2 \mathbb{Z}_0 e^{i\omega t} &= -\frac{1}{m} \mathbb{K} \mathbb{Z}_0 e^{i\omega t} \\ m\omega^2 \mathbb{Z}_0 &= \mathbb{K} \mathbb{Z}_0\end{aligned}$$

This is an eigenvalue (or eigenmode) equation.

Note that we have different  $\omega$  for each normal mode.

$$\begin{aligned}x_+ \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \omega_+ &= \sqrt{\frac{k}{m}} \\ x_- \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \omega_- &= \sqrt{\frac{k+2k'}{m}}\end{aligned}$$

We can solve the eigenvalue problem

$$\begin{aligned}\mathbb{K} \mathbb{Z}_0 - m\omega^2 \mathbb{Z}_0 &= 0 \\ \underbrace{(\mathbb{K} - m\omega^2 \mathbb{I})}_{\mathbb{A}} \mathbb{Z}_0 &= 0\end{aligned}$$

This is of the form  $\mathbb{A} \mathbb{Z}_0 = 0$

We can solve this with

$$\begin{aligned}\mathbb{A}^{-1} \mathbb{A} \mathbb{Z}_0 &= \mathbb{A}^{-1} 0 \\ \mathbb{Z}_0 &= \begin{pmatrix} 0 \\ 0 \end{pmatrix}\end{aligned}$$

We need a non invertible matrix

$$\det(\mathbb{K} - m\omega^2 \mathbb{I}) = 0$$

Lets find the eigenvalues

$$\begin{pmatrix} k + k' - m\omega^2 & -k' \\ -k' & k + k' - m\omega^2 \end{pmatrix} = 0$$

Taking the determinant yields

$$(k + k' - m\omega^2)^2 - k'^2 = 0$$

This is a difference of two squares, in the form  $(a + b)(a - b)$

$$(k + k' - m\omega^2 + k')(k + k' - m\omega^2 - k') = 0$$

This yields two different solutions for what  $\omega$  can be (which should agree with what we found before)!

$$\begin{aligned}m\omega^2 &= k + 2k' \\ \omega_1 &= \sqrt{\frac{k + 2k'}{m}} = \omega_-\end{aligned}$$

We can plug in the values of  $\omega$  to find for our eigenvectors.

For example

$$\begin{aligned}(\mathbb{K} - m\omega^2 \mathbb{I}) \begin{pmatrix} a \\ b \end{pmatrix} &= 0 \\ \begin{pmatrix} k + k' - \frac{m(k+2k')}{m} & -k' \\ -k' & k + k' - m\omega^2 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} &= \begin{pmatrix} 0 \\ 0 \end{pmatrix} \\ k' \begin{pmatrix} -1 & -1 \\ -1 & -1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} &= \begin{pmatrix} 0 \\ 0 \end{pmatrix}\end{aligned}$$

This gives us the equations

$$-a - b = 0, \text{ so } a = -b.$$

This is our eigenvector,  $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$

You can do the same thing with the other normal mode for  $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ .

We now have the complex functions of time with any complex coefficients  $A_1, A_2$  as

$$\begin{aligned} \mathbb{Z}(t) &= A_1 \mathbb{Z}_0^1 e^{i\omega_1 t} + A_2 \mathbb{Z}_0^2 e^{i\omega_2 t} \\ &\quad \text{where here,} \\ \mathbb{Z}_0^1 &= \frac{1}{2} \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \omega_1 = \sqrt{\frac{k + 2k'}{m}} \\ \mathbb{Z}_0^2 &= \frac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \omega_2 = \sqrt{\frac{k}{m}} \end{aligned}$$

(note that the  $\frac{1}{2}$  here is arbitrary, but we write it to match )

## Practice problem

Lets take three masses with 4 springs between them, all of mass  $m$  and spring constant  $k$ .

The legrangian for this is

$$\begin{aligned} U &= \frac{1}{2}k(x_3^2 + (x_3 - x_2)^2 + (x_2 - x_1)^2 + x_1^2) \\ T &= \frac{1}{2}m(\dot{x}_1^2 + \dot{x}_2^2 + \dot{x}_3^2) \\ \mathcal{L} &= \frac{1}{2}m(\dot{x}_1^2 + \dot{x}_2^2 + \dot{x}_3^2) - \frac{1}{2}k(x_3^2 + (x_3 - x_2)^2 + (x_2 - x_1)^2 + x_1^2) \end{aligned}$$

We can find the EL equations for these

$$\begin{aligned} m\ddot{x}_1 &= \frac{1}{2}k(2x_1 - 2x_2 + 2x_1) \\ m\ddot{x}_2 &= \frac{1}{2}k(2x_2 - 2x_3) \\ m\ddot{x}_3 &= \frac{1}{2}k(2x_3 + 2x_3 - 2x_2) \end{aligned}$$

We can write these as a matrix form

$$\begin{pmatrix} \ddot{x}_1 \\ \ddot{x}_2 \\ \ddot{x}_3 \end{pmatrix} = \frac{k}{2m} \begin{pmatrix} 4 & -2 & 0 \\ 0 & 2 & -2 \\ 0 & -2 & 4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

We can take this as the matrix equation

$$\ddot{\mathbb{X}} = \frac{k}{2m} \mathbb{K}$$